Stochastic Calculus

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Outline

Module Goals

Introduction
- Financial Instruments

Probability
- Random Variables and Distributions
- Expected Value and Variance

Stochastic Process
- Binomial Model
- Option Pricing Model
- Random Walks
- Continuous Model
- Ito Integral

Geometric Brownian Motion

Black and Scholes
- Ejercicio Renta Fija

- Understand the concept of uncertainty and its relationship with measurement methods.
- Understand the concept of random walks and Brownian motionss.
- Recognize a Markovian process and apply to discrete and continuous stochastic processes.
- Associate the fundamental concepts of the integral Ito to stochastic processes
- Use stochastic processes to model financial time series and other problems related to corporate finance.

Presentación

- Contenido
- Evaluación
  - 2 reseñas individuales. (30 %)
  - 2 talleres (25 %)
  - 2 quices (20 %)
  - Proyecto final 25 %
- Horario de Atención
- Proyecto Final

Stochastic models are the basis for forecasting, monitoring and analyzing investment projects. They emulate different scenarios that include uncertainty.

- What is uncertainty?
- What is Risk?
- How can I measure risk?

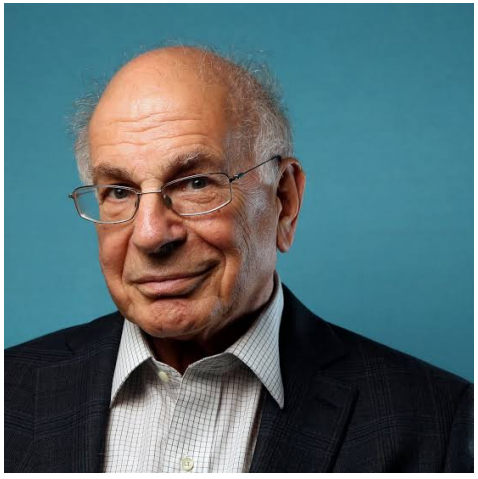
Financial context to understand and apply the mathematical models used for the management, monitoring and administration of financial risk in investment projects.



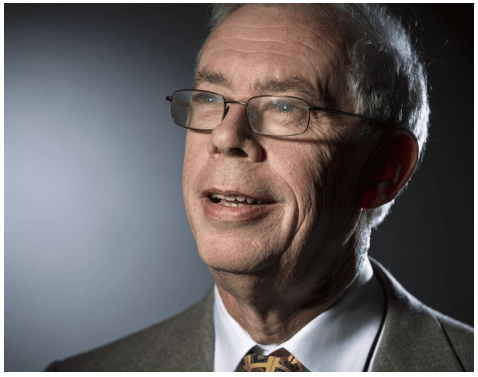
Warren Buffet



Jim Simons



Daniel Kahneman



John Kay

Boundary Rationality and Boundary Cognitive

- Can we maximize a profit?
- Do we have all the information?
- what we want to optimize?

What is the strategy?

- Make a prediction
- Make a prescription

We produce information to make decision

- What is Foresight?
- What is Forecasting?

Probability to have losses of the expected returns.

Risk Factor

- Log Returns of stocks
- The interest rate
- The exchange rate
- Implied volatility

Underlying Financial Instruments

Stocks		**Bonds**	
Standard		Standard	
Dividens		Cupon	
Share		Debt	
Price $t = 0$	S_0	Maturity Price	$B(T,T) = 1$
Price t	S_t	Price $t = 0$	$B(0,T) = e^{-rT}$
Mean Returns	μ	Price t	$B(t,T) = e^{-r(T-t)}$
Variance Returns	σ^2	Interest Rate	r
Without Expiration		Have Expiration	T

Derivatives: Were introduced by the ancient cultures in the trade and hedge of commodities (agriculture products) through the study of the astronomy after predict the rain and dry seasons.

Can be defined as a financial instrument whose value depends from the values of other underlying variables.

Today there are trade in standardizied markets and in OTC (over the count) markets

Type of investors

1. Hedge Instruments
2. Speculative
3. Arbitrage

Some derivatives instruments [Hull, 2003]

1. Forwards and Futures
2. Options
3. Swaps

Most important trade centers

1. CBOE Chicago board options exchange
2. CME Chicago mercantile Exchange
3. LME London Metal Exchange
4. Bolsa mercantil de Colombia

Definition: Let Ω be a nonempty set, and let $\mathcal{F}$ be a collection of subsets on Ω. We say that $\mathcal{F}$ is a σ [Shreve, 2004] :

- The empty set $\emptyset$ belongs to $\mathcal{F}$
- whenever a set $\mathcal{A}$ belongs to $\mathcal{F}$, $\mathcal{A}^c$ also belongs to $\mathcal{F}$.
- whenever a sequence of sets $\mathcal{A}_1, \mathcal{A}_2, ...\mathcal{A}_n$ belongs to $\mathcal{F}$, their union $\cup$ of $\mathcal{A}_1, \mathcal{A}_2, ...\mathcal{A}_n$ also belongs to $\mathcal{F}$.

Definition Let Ω be a nonempty set with a collection of subsets on $\mathcal{F}$. A probability measure $\mathbb{P}$ is a function that, to every set $Ain\mathcal{F}$ assigns a number $[0, 1]$ called the probability of A and written $\mathbb{P}(A)$ [Shreve, 2004]:

- $\mathbb{P}(\Omega) = 1$
- whenever $A_1, A_2, ...A_n$ is a sequence of disjoint sets in $\mathcal{F}$
- $$\mathbb{P}\big(\cup_{n=1}^{\infty} \mathcal{A}_n\big) = \sum_{n=1}^{\infty} \mathbb{P}(\mathcal{A}_n)$$

A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$

Theorem: Let $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space. For A, B $(A_n)_{n\in\mathbb{N}}$ y $(B_n)_{n\in\mathbb{N}} \in \mathcal{F}$

- $\mathbb{P}(\emptyset) = 0$
- $P(A^c) = 1 - P(A)$
- Si $A \subseteq B$ then $P(A) \leq P(B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Example: We toss a coin infinitely many times and let Ω denote the set of possible outcomes. We assume the probability of head on each toss is $p > 0$, the probability of a tail is $q = 1 - p > 0$ and the different tosses are independent. Construct the outcomes for $\mathcal{F}$ from 0 to 2.

Definition: Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A random variable is a real valued function X defined on Ω with the property that for every Borel subset B of $\mathbb{R}$, the subset of Ω given by [Shreve, 2012]

$$\{X \in B\} = \{\omega \in \Omega; X(\omega) \in B\}$$

is in the σ algebra $\mathcal{F}$.

Definition: Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The distribution measure of X is the probability measure μ_x that assigns to each Borel subset B of $\mathbb{R}$ the mass $\mu_X(B) = P\{X \in B\}$

Expected Value and Variance

Definition Let $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space and X a random variable [Shreve, 2012].

a If X is a discrete random variable $x_1, x_2, ...$ the expected value of X exits if $\sum_{k=1}^{\infty} |x_k| = P(\{X = x_k\}) < \infty$.
Then

$$\mathbb{E}(X) = \sum_x x\mathbb{P}(x) \tag{1}$$

b If X is a continuous random variable with density function $f(x)$, the expected value for X is $\int_{-\infty}^{\infty} |x| f(x)dx < \infty$

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x)dx \tag{2}$$

c If $E[X] < \infty$, then $Var[X] := E([X - E(X)]^2)$ a
$\sigma = \sqrt{Var(X)}$

Expected Value: Let $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space and X a random variable

- $E(a) = a$ for all $a \in \mathbb{R}$.
- if $X_1 \leq X_2$, then $E(X_1) \leq E(X_2)$
- $E(aX_1 + bX_2) = aE(X_1) + bE(X_2)$ for all $a, b \in \mathbb{R}$
- $|E(X)| \leq E(|X|)$

Variance Let $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space and X a random variable.

- $Var(X) = E(X^2) - (E(X))^2$
- $Var(aX + b) = a^2 Var(X)$ para $a, b \in \mathbb{R}$

Binomial Distribution: Let x a discrete random variable.
$x \sim B(n, p)$

$$P(X = x) = \binom{n}{x} p^x q^{n-x} \tag{3}$$

With p and q as probability in $(0, 1)$ and n as number of attempts

Normal Distribution Let x a continuous random variable $x \sim N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

With μ as mean and σ^2 as variance

Theorem: Let $(\Omega, \mathcal{F}, \mathbb{P})$ as a probability space, and $\mathcal{G}$ is a subspace of $\mathcal{F}$. [Shreve, 2004]

- Linearity: If X and Y are continuous variables and a and b are constants, then:

$$\mathbb{E}[aX + bY|\mathcal{G}] = a\mathbb{E}[X|\mathcal{G}] + b\mathbb{E}[Y|\mathcal{G}]$$

- Get what is known: If X and Y are continuous variables and XY is continuous and X is measurable in $\mathcal{G}$, then:

$$\mathbb{E}[XY|\mathcal{G}] = X\mathbb{E}[Y|\mathcal{G}]$$

- Iterated Condition: If $\mathcal{H}$ is a σ algebra of $\mathcal{G}$ and X is a continuous variable, then

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}|\mathcal{H}] = \mathbb{E}[X|\mathcal{H}]$$

- If X is continuous and independent of $\mathcal{G}$, then

$$\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$$

A stochastic process is a collection of random variables indexed by time. It will be useful to consider separately the cases of discrete time and continuous time. There are two kind of stochastic process. First a discrete time stochastic process $X = \{X_n, n = 0, 1, 2, ...\}$ is a countable collection of random variables indexed by the non-negative integers, and a continuous time stochastic process $X = \{X_t, 0 \leq t < \infty\}$ is an uncountable collection of random variables indexed by the non-negative real numbers.

Key Features

Arbitrage: One of the fundamental concepts in the theory of option pricing is the absence of arbitrage opportunities, is called the no arbitrage principle.

A complete market should have at least three conditions to full the no arbitrage principle

a Depth in transactions

b Liquidity

c The same information for all participants

More precisely, an arbitrage opportunity can be defined as a self-financing trading strategy requiring no initial investment, having zero probability of negative value at expiration. and vet having some possibility of a positive terminal payoff.

Neutral Risk $\mathbb{Q}$: A probability measure in $[0, 1]$ is equivalent to $\mathbb{P}$, but with the effects from a financial random variable in a complete market.

In other words, an investor does not control the market and he can only measure the tendency and the variance to take a position and make a strategy without the certain to get benefits from the market.

No arbitrage opportunities exist if and only if there exists a risk neutral probability measure $\mathcal{Q}$.

Example: The probability of the movement of an asset up is $\widetilde{p}$ or down $\widetilde{q} = 1 - \widetilde{p}$. When the asset foes up the magnitude is $U = e^{\sigma\sqrt{t}}$ and when the asset goes down the magnitude is $D = e^{-\sigma\sqrt{t}}$. Using the risk free rate r, a risk neutral measure is:

$$\widetilde{p} = \frac{(1+r) - D}{U - D} \tag{4}$$

It is only true for 4 when

$$D < (1+r) < U \tag{5}$$

Filtration Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space on which is defined a stochastic process. A filtration is a collector of σ algebra $\mathcal{F}(t)$

i For $0 \leq s \leq t$ every set in $\mathcal{F}(s)$ is also in $\mathcal{F}(t)$. In other words there is at least as much information available at the later time $\mathcal{F}(t)$ as there is at the earlier time $\mathcal{F}(s)$.

ii For each $t \geq 0$ a stochastic process at the time t is $\mathcal{F}(t)$ measurable. In other words, the information available at the time t is sufficient to evaluate the stochastic process.

iii For $0 \leq t < u$, the increment $X(u) - X(t)$ is independent of $\mathcal{F}(t)$.

Definition A markov is a process in a space $(\Omega, \mathcal{F}, \mathbb{P})$ with a filtration $\mathcal{F}_{k=0}^{n}$ on an event $\mathcal{F}$ and a stochastic process X_t. Implementing a binomial model, for example, an adapted process $X_0, X_1, ..., X_n$ is Markovian if for each n between 0 and $N-1$ and for each function $f(x)$ there is another function $g(x)$ that depend on n and f [Shreve, 2012].

$$\mathbb{E}_n[f(X_{n+1}] = g(X_n) \tag{6}$$

This means that 6 is random and depends on the first n coin flips, *The property says its dependency on coin flips occurs through* X_n[Shreve, 2012].

Martingale Property Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, Let T be a fixed positive number, and let $\mathcal{F}(t)$, $0 \leq t \leq T$ be a filtration of sub σ algebra of $\mathcal{F}$. Consider an adapted stochastic process $M(t), \quad 0 \leq s \leq t \leq T$.[Shreve, 2004]

i If $\mathbb{E}[M(t)|\mathcal{F}(s)] = M(s)$ for all $0 \leq s \leq t \leq T$ then $M(t)$ is a Martingale and has no tendency to rise or fall.

ii If $\mathbb{E}[M(t)|\mathcal{F}(s)] \geq M(s)$ for all $0 \leq s \leq t \leq T$ then $M(t)$ is a sub martingale and it may have tendency to rise.

iii If $\mathbb{E}[M(t)|\mathcal{F}(s)] \leq M(s)$ for all $0 \leq s \leq t \leq T$ then $M(t)$ is a super martingale and it may have a tendency to fall.

The future value is the same as the present value when exist a risk neutral measure $\mathbb{Q}$ and under a risk free rate it is also a martingale.

$$\frac{S_n}{(1+r)^n} = \mathbb{E}_n^{\mathbb{Q}}\left[\frac{S_{n+1}}{(1+r)^{n+1}}\right] \tag{7}$$

Using 5 the probability measure $\widetilde{q} = (1 - \widetilde{p})$ and $D = 1/U$.

$$\mathbb{E}_n^{\mathbb{Q}}\left[\frac{S_{n+1}}{(1+r)^{n+1}}\right] = \mathbb{E}_n^{\mathbb{Q}}\left[\frac{S_n}{(1+r)^{n+1}}\frac{S_{n+1}}{S_n}\right] \tag{8}$$

$$= \frac{S_n}{(1+r)^n}\mathbb{E}_n^{\mathbb{Q}}\left[\frac{1}{(1+r)}\frac{S_{n+1}}{S_n}\right] \tag{9}$$

$$\frac{S_n}{(1+r)^n}\frac{1}{(1+r)}\mathbb{E}_n^{\mathbb{Q}}\left[\frac{S_{n+1}}{S_n}\right] \tag{10}$$

$$\frac{S_n}{(1+r)^n}\frac{\widetilde{p}U + \widetilde{q}D}{1+r} \tag{11}$$

$$\frac{S_n}{(1+r)^n} \tag{12}$$

Stock prices are assumed to follow this simple model: The initial stock price during the period under study is denoted S_0.

- The stock price moves up by a factor U if the coin comes out heads H.
- Down by a factor of D if it comes out tails T

$$\Omega = \{U, D\}^n = \{(\omega_1, \omega_2, ..., \omega_n\}; \omega_i = U \quad o \quad \omega_i = D$$

$$X_i = \Omega \to \mathbb{R} \quad \omega \to \omega_i$$

$$H_i : \Omega \to \mathbb{R} \quad \omega \to \#\{j \leq i : \omega_j = U\}$$

$$T_i : \Omega \to \mathbb{R} \quad \omega \to \#\{j \leq i : \omega_j = D\}$$

The stock price at the moment S_i is compute by the multiplication of the results of the random variable with S_0.

$$S_i = S_0 \Pi_{j=1}^{i} X_j = S_0 U^{H_i} D^{T_i} \tag{13}$$

The expected value is the sum from each of the results multiplied by its probability.

$$\mathbb{E}_{\mathbb{Q}} \left(\frac{1}{(1+R)^j} S_j | \mathcal{F}_i \right) = \frac{1}{(1+R)^i} S_i \quad para \quad i < j$$

Where the filtration $\mathcal{F}_i$ $i \geq 0$ is a martingale for $0 \leq i \leq j \leq n$

$$S_j = S_i \Pi_{k=i+1}^{j} X_k$$

Let $\mathcal{F}_i$ adapted, $\Pi_{k=i+1}^{j} X_k$ and independent from $\mathcal{F}_i$

$$\mathbb{E}_{\mathbb{Q}}(S_j|\mathcal{F}_i) = S_i \mathbb{E}_{\mathbb{Q}}(\Pi_{k=i+1}^{j} X_k|\mathcal{F}_i)$$

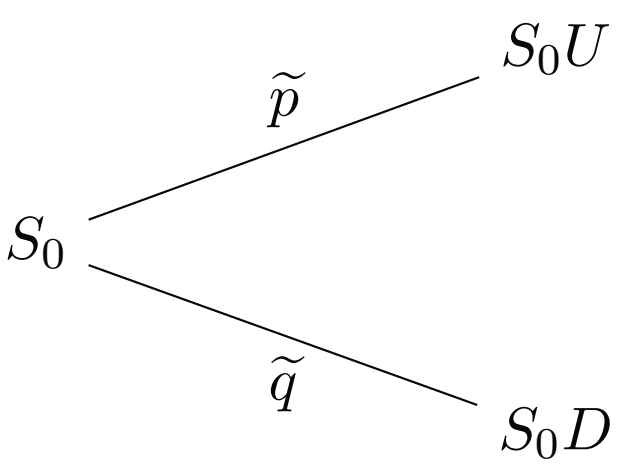
$$= S_i \mathbb{E}_{\mathbb{Q}}(\Pi_{k=i+1}^{j} X_k) = S_i[U\widetilde{p} + D\widetilde{q}]^{j-i}$$

$$= S_i(1+R)^{j-i}$$

Using 13 and 3 you can get the expected value of S_n for any $\mathcal{F}_n$ in a discrete time $(0, n)$ [Cox et al., 1979].

$$E_{\mathbb{Q}}[S_n|\mathcal{F}_n] = \sum_{x=0}^{n} \binom{n}{x} \widetilde{p}^x (1-\widetilde{p})^{n-x} S_0 U^x D^{n-x} \tag{14}$$

The risk neutral probability is the same as 4.

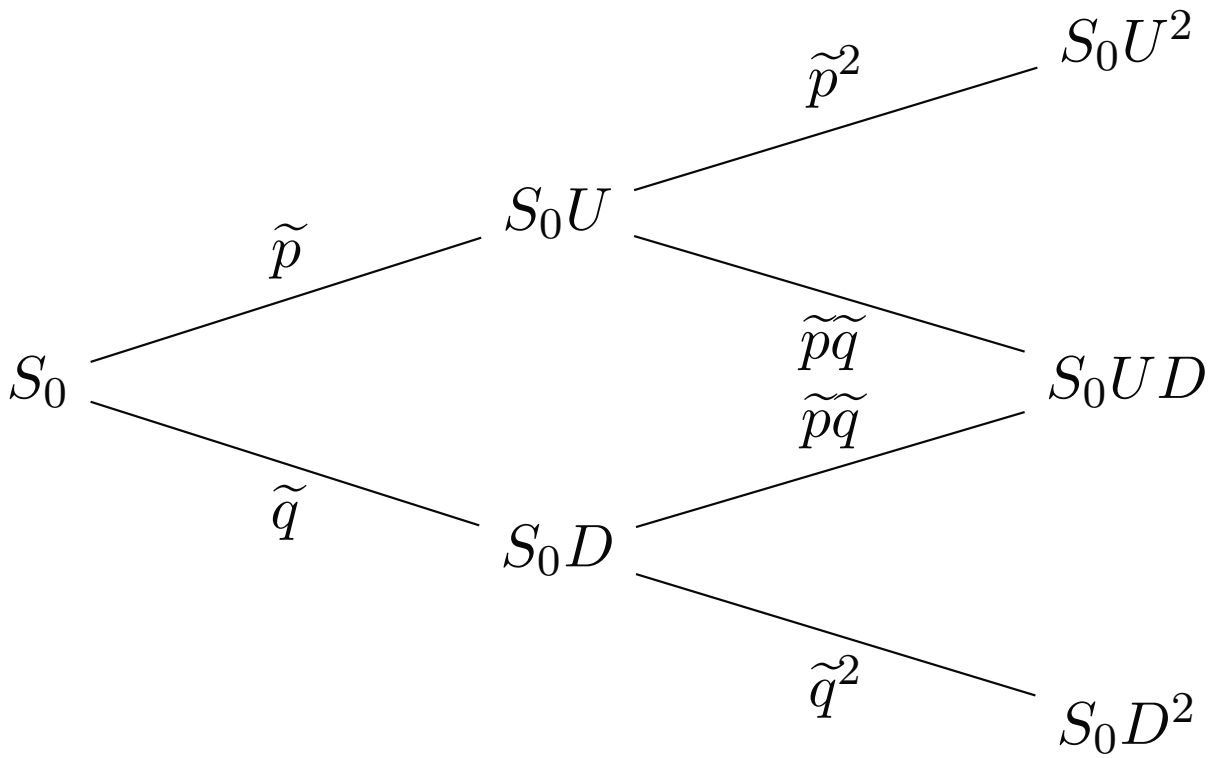


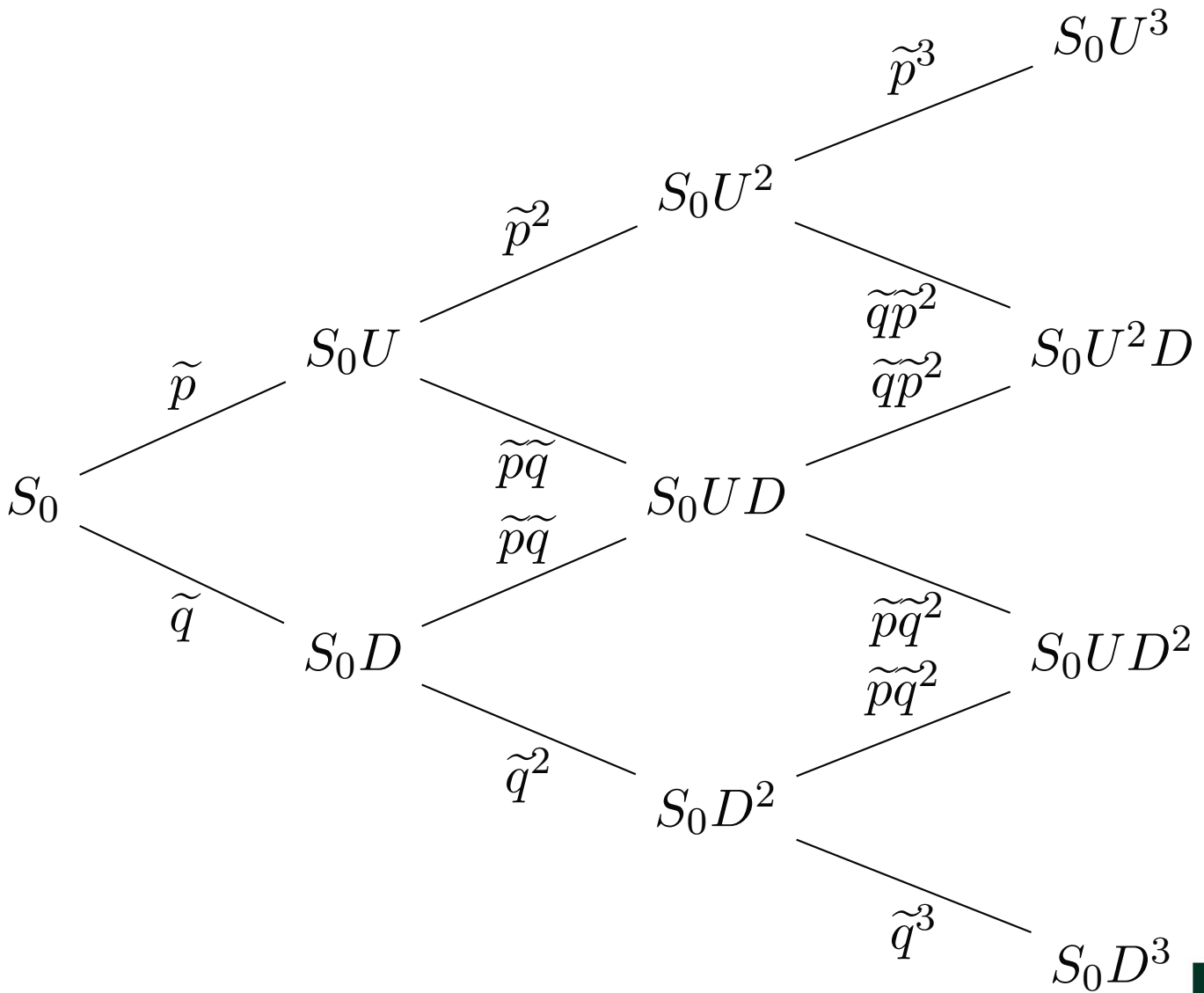
$$S_0$$

$$\widetilde{q} = 1 - \widetilde{p}$$

$$U = e^{\sigma\sqrt{dt}}$$

$$D = e^{-\sigma\sqrt{dt}}$$


S_0
$\widetilde{p}$
$\widetilde{q}$
S_0U
S_0D
$\widetilde{p}^2$
$\widetilde{p}\widetilde{q}$
$\widetilde{p}\widetilde{q}$
$\widetilde{q}^2$
S_0U^2
S_0UD
S_0D^2


S_0
$\widetilde{p}$
$\widetilde{q}$
S_0U
S_0D
$\widetilde{p}^2$
$\widetilde{p}\widetilde{q}$
$\widetilde{p}\widetilde{q}$
$\widetilde{q}^2$
S_0U^2
S_0UD
S_0D^2
$\widetilde{p}^3$
$\widetilde{q}\widetilde{p}^2$
$\widetilde{q}\widetilde{p}^2$
$\widetilde{p}\widetilde{q}^2$
$\widetilde{p}\widetilde{q}^2$
$\widetilde{q}^3$
S_0U^3
S_0U^2D
S_0UD^2
S_0D^3

Model Cox Ross Rubinstein The problem now is to find the exact formula or method which transform the asset price and the time of the expiration into the value of an option.
[Cox et al., 1979]. Using the price function and the option payoff there is a function to get the current option value for an European/american option.

$$S_n = S_0 U^x D^{n-x} \tag{15}$$

For each option there is a payoff function.

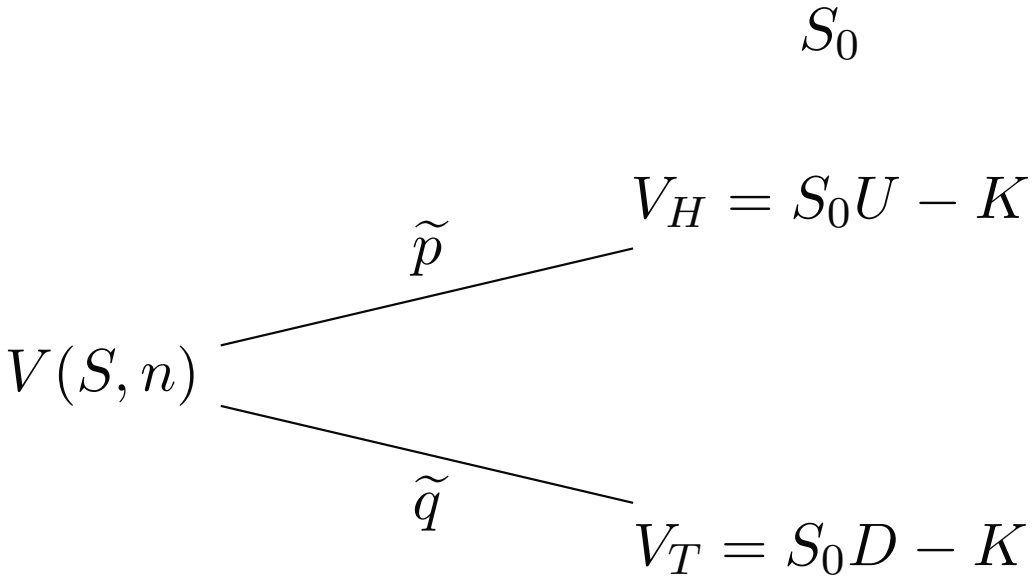
- Option Call (buy) $V(S,n) = max(S-K,0)$.
- Option Put (sell) $V(S,n) = max(K-S,0)$

There is an underlying price S and a strike price K.

$$V(S,n) = \left(\sum_{x=0}^{n} \binom{n}{x} \widetilde{p}^{x} (1-\widetilde{p})^{n-x} max(S_0 U^x D^{n-x} - K, 0) \right) / (1+r)^n \tag{16}$$

$$V(S,n) = \left(\sum_{x=0}^{n} \binom{n}{x} \widetilde{p}^{x} (1-\widetilde{p})^{n-x} max(K - S_0 U^x D^{n-x}, 0) \right) / (1+r)^n$$

One Step Model



K=strike

One Step Model:Suppose a derivative security pays off the amount V_1 at the time 1, where V_1 is a $\mathcal{F}_1$ measurable random variable. We construct a self- financing portfolio.

- Sell the security for V_0 at time 0
- Buy Δ shares of stock at time 0
- Invest $V_0 - \Delta S_0$ in the money market, at risk free rate r

The wealth at time 1 is

$$V_1 = (1+r)V_0 + \Delta(S(1) - (1+r)S_0) \tag{18}$$

It could be negative

$$\begin{aligned} V_1(H) &= (1+r)V_0 + \Delta(S_1(H) - (1+r)S_0) \\ V_1(T) &= (1+r)V_0 + \Delta(S_1(T) - (1+r)S_0) \end{aligned}$$

Finding Δ

$$\Delta = \frac{V_1(H) - V_1(T)}{S_1(H) - S_1(T)},$$

$$V_1(H) = (1+r)V_0 + \Delta(S_1(H) - (1+r)S_0) \tag{19}$$

$$V_1(T) = (1+r)V_0 + \Delta(S_1(T) - (1+r)S_0) \tag{20}$$

Finding Δ

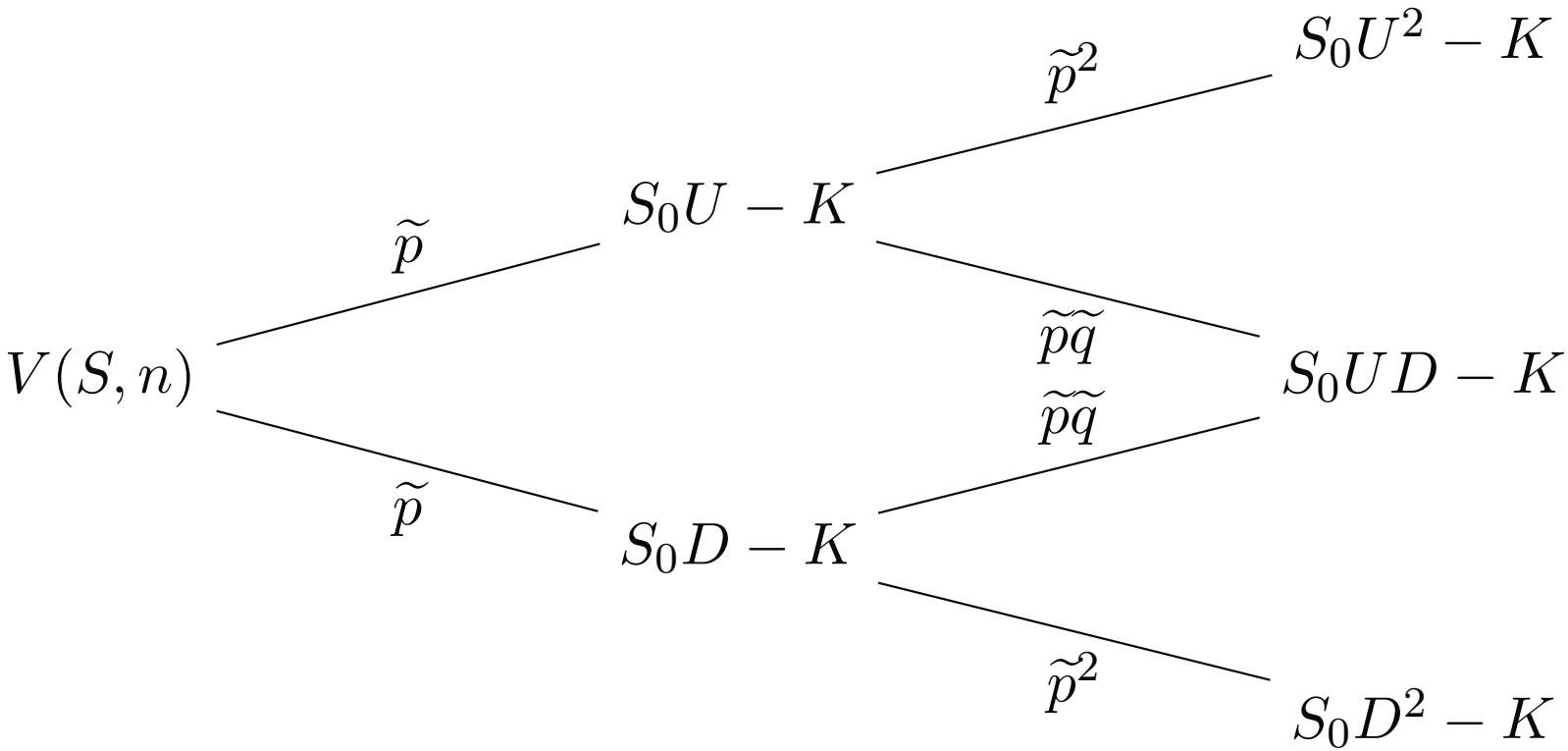
$$\Delta = \frac{V_1(H) - V_1(T)}{S_1(H) - S_1(T)}, \tag{21}$$

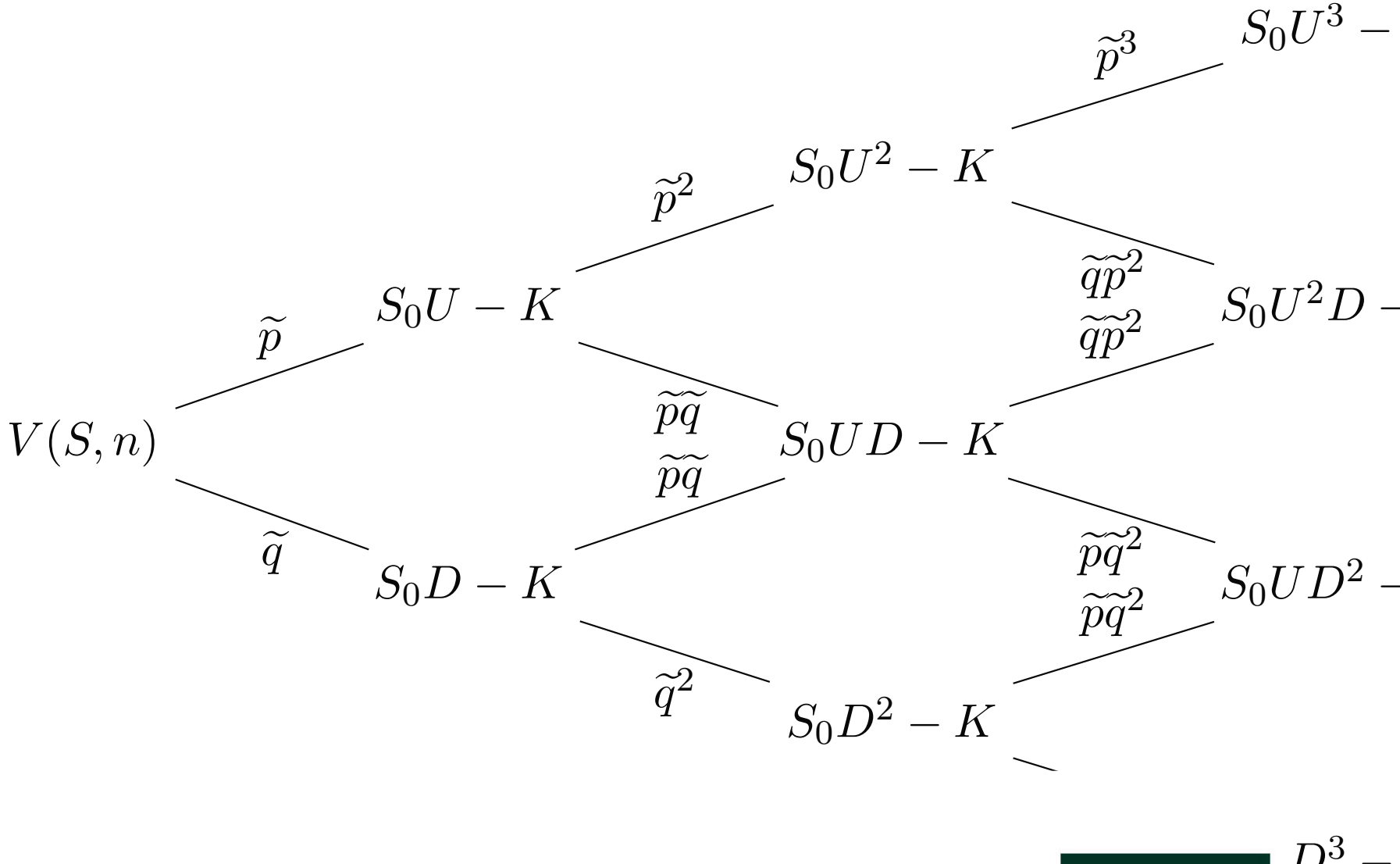
Plug 21 in 19

$$\begin{aligned}
(1+r)V_0 &= V_1(H) - \Delta(S_1(H) - (1+r)S_0) \\
&= V_1(H) - \tfrac{V_1(H)-V_1(T)}{(U-D)S_0}(\Delta(S_1(H) - (1+r)S_0) \\
&\quad (U-1-r)S_0 \\
&= \tfrac{1}{U-D}[(U-D)V_1(H) - (V_1(H) - V_1(T))(U-1-r)] \\
&= \tfrac{(1+r)-D}{U-D}V_1(H) + \tfrac{U-(1+r)}{U-D}V_1(T)
\end{aligned}$$

$$\widetilde{p} = \frac{(1+r)-D}{(U-D)}, \quad \widetilde{q} = \frac{U-(1+r)}{(U-D)}$$

$$V_0 = \frac{1}{1+r}[\widetilde{p}V(H) + \widetilde{q}V(T)] \qquad (22)$$


$V(S,n)$
$\widetilde{p}$
$\widetilde{p}$
$S_0U - K$
$S_0D - K$
$\widetilde{p}^2$
$\widetilde{p}\widetilde{q}$
$\widetilde{p}\widetilde{q}$
$\widetilde{p}^2$
$S_0U^2 - K$
$S_0UD - K$
$S_0D^2 - K$


$V(S,n)$
$\widetilde{p}$
$\widetilde{q}$
$S_0U - K$
$S_0D - K$
$\widetilde{p}^2$
$\widetilde{p}\widetilde{q}$
$\widetilde{p}\widetilde{q}$
$\widetilde{q}^2$
$S_0U^2 - K$
$S_0UD - K$
$S_0D^2 - K$
$\widetilde{p}^3$
$\widetilde{q}\widetilde{p}^2$
$\widetilde{q}\widetilde{p}^2$
$\widetilde{p}\widetilde{q}^2$
$\widetilde{p}\widetilde{q}^2$
$S_0U^3 -$
$S_0U^2D -$
$S_0UD^2 -$
$D^3 -$

A symmetric random walk is a movement that arises from the exercise of tossing a coin with a probability of $\frac{1}{2}$, n times. Your results can be:

$$X_j = \begin{cases} 1 & \text{Si} \quad \omega_j = H \\ -1 & \text{Si} \quad \omega_j = T \end{cases}$$

The distance between each result is the difference of a random variable $X_n - X_{n-1} = \Delta X_n$ where the filter $\mathcal{F}_{n\geq 0}$ allows to obtain the parameters of the distribution that must be iid (independent, and identically distributed) so that the distance of a random variable in a walk is defined as $\Delta X_n \sim iid$.

With an initial value $M_0 = 0$ and as the sum of many random variables X_n, $M_n = \sum_{j=1}^{n} X_j, n = 1, 2...$ The creation of a random walk is a step by step where the future value M_n depends on the previous value M_{n-1} plus the distance ΔM_n

$$M_n = M_{n-1} + \Delta M_n = M_n - M_{n-1} = \Delta M_n$$

$$M_{n+1} = M_n + \Delta M_{n+1} \tag{23}$$

The symmetric probability $P = 0{,}5$ gives the value for X between1 and -1 and the expected value of a random variable X applying 1 we obtain that $E[X] = 1 * 0{,}5 + -1 * 0{,}5 = 0$ while the variance is equal to $Var[X] = 1$ using the properties of variance.
In the same logic a random walk M_n using its independent increments, we have that they are the sum of the variable X_j so far n its expected value depends on the expected value of X which is zero.

$$M_{k_1} = (M_{k_1} - M_{k_0}), (M_{k_2} - M_{k_1}), ..., (M_{k_m} - M_{k_{m-1}})$$

where $M_{k_{i+1}} - M_{k_i} = \sum_{j=k_{i+1}}^{k_{i+1}} X_j$

$$E[M_{k_{i+1}} - M_{k_i}] = 0$$

While the variance is the sum of all the variances of X which in this case results in n

$$Var[M_{k_{i+1}} - M_{k_i}] = \sum_{j=k_i+1}^{k_{i+1}} = Var(X_j) = \sum_{j=k_i+1}^{k_{i+1}} 1 = k_{i+1} - k_i \tag{25}$$

Martingale of a Random Walk: Given a symmetric random walk with a filtration $\mathcal{F}_{n\geq 0}$ and $m < n$ the expected value can be obtained conditional on the information of the moment n as follows:
[Shreve, 2004]:

$$\begin{aligned}
\mathbb{E}[M_n|\mathcal{F}_m] &= \mathbb{E}[(M_n - M_m) + M_m|\mathcal{F}_m] \\
&= \mathbb{E}[(M_n - M_m)|\mathcal{F}_m] + \mathbb{E}[M_m|\mathcal{F}_m] \\
&= \mathbb{E}[(M_n - M_m)|\mathcal{F}_m] + M_m \\
&= \mathbb{E}[(M_n - M_m)] + M_m \\
&= M_m
\end{aligned}$$

Example: A coin is tossed several times. Assume the probability of $1/2$ for heads or tails. Being $X_j = 1$ for face and $X_j = -1$ for stamp. Consider the stochastic process $M_0, M_1, ...$ with $M_0 = 0$ and $M_n = \sum_{j=1}^{n} X_j$ for $n \geq 1$.

a Is it M_n a martingale?

b Let be σ positiv for $n \geq 0$ ¿es S_n a martingale for the next process.

$$S_n = e^{\sigma M_n}\left(\frac{2}{e^{\sigma} + e^{-\sigma}}\right)$$

$$\begin{aligned} E_n(M_{n+1}) &= \sum_{j=1}^{n+1} E_n[X_j] \\ &= E_n[X_{n+1}] + \sum_{j=1}^{n} X_j \\ &= 0 + M_n \end{aligned}$$

$$
\begin{aligned}
S_n &= e^{\sigma M_n}\left(\frac{2}{e^{\sigma}+e^{-\sigma}}\right) \\
S_{n+1} &= S_n e^{\sigma M_n}\left(\frac{2}{e^{\sigma}+e^{-\sigma}}\right) \\
\mathbb{E}[S_{n+1}] &= S_n \mathbb{E} e^{\sigma X_{n+1}}\left(\frac{2}{e^{\sigma}+e^{-\sigma}}\right) \\
\mathbb{E}[S_{n+1}] &= S_n\left(\frac{1}{2}e^{\sigma}+\frac{1}{2}e^{-\sigma}\right)\left(\frac{2}{e^{\sigma}+e^{-\sigma}}\right) \\
\mathbb{E}[S_{n+1}] &= S_n
\end{aligned}
$$

The approximation to a Brownian motion is obtained with the help of a symmetric random walk where the probability of $p = \frac{1}{2}$[Shreve, 2004]

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} M_{nt} \tag{26}$$

$$\mathbb{E}[W^{(n)}(t) - W^{(n)}(s)] = 0 \tag{27}$$

$$Var[W^{(n)}(t) - W^{(n)}(s)] = t - s \tag{28}$$

The martingale for $\mathbb{E}[W^n(t)|\mathcal{F}(s)] = W^n(s)$ for $0 \leq s \leq t$.
Quadratic variation

$$[W^n, W^n](t) = \sum_{j=1}^{nt} \left[W^n\left(\frac{j}{n}\right) - W^n\left(\frac{j-1}{n}\right) \right]^2$$

$$= \sum_{j=1}^{nt} \left[\frac{1}{\sqrt{n}} X_j \right]^2 = \sum_{j=1}^{nt} \frac{1}{n} = t$$


Random Walk
M(t)
t
20
10
0
−10
−20
−30
0
20
40
60
80

For $t \geq 0$, if $n \to \infty$ the distribution of a scaled random walk $W^n(t)$ evaluated at t converges towards a normal distribution with mean 0 and variance t.[Shreve, 2004]

The price of a stock is distributed in a Log-Normal way, using the properties of the binomial model, the limit can be obtained by showing that the price has from a binomial distribution to a Log Normal distribution.

Using nt as an integer to scale the process, reformulating $U_n = 1 + \frac{\sigma}{\sqrt{n}}$ and $D_n = 1 - \frac{\sigma}{\sqrt{n}}$ a new risk neutral probability can be obtained.

$$\widetilde{p} = \frac{(1+r) - D_n}{U_n - D_n} = \frac{\sigma/\sqrt{n}}{2\sigma/\sqrt{n}} = \frac{1}{2}, \quad \widetilde{q} = \frac{U_n - (1+r)}{U_n - D_n} = \frac{\sigma/\sqrt{n}}{2\sigma/\sqrt{n}} = \frac{1}{2}$$

$$nt = H_{nt} + T_{nt}$$

$$M_{nt} = H_{nt} - T_{nt}$$

$$H_{nt} = \frac{1}{2}(nt + M_{nt}) \quad T_{nt} = \frac{1}{2}(nt - M_{nt})$$

$$S_n(t) = S(0)U_n^{H_{nt}} D_n^{T_{nt}} = S(0)\left(1+\frac{\sigma}{\sqrt{n}}\right)^{\frac{1}{2}(nt+M_{nt})}\left(1-\frac{\sigma}{\sqrt{n}}\right)^{\frac{1}{2}(nt-M_{nt})}$$

$$S(t) = S(0)exp\left\{\sigma W(t) - \frac{1}{2}\sigma^2 t\right\} \tag{29}$$

Where $W(t)$ has mean 0 and variance t.

$$\begin{aligned} logS_n(t) &= logS(0) + \tfrac{1}{2}(nt + M_{nt})log\left(1 + \tfrac{\sigma}{\sqrt{n}}\right) \\ &\quad + \tfrac{1}{2}(nt - M_{nt})log\left(1 - \tfrac{\sigma}{\sqrt{n}}\right) \\ logS(t) &= logS(0) + \sigma W(t) - \tfrac{1}{2}\sigma^2 t \end{aligned} \tag{30}$$

The Taylor series derives the function with respect to the variables on which it depends until a value of zero is obtained. For example the function $f(x) = log(1+x)$ is derived as $f'(x) = (1+x)^{-1}$ y $f''(x) = -(1+x)^{-2}$ and evaluated at zero we get $f'(0) = 1$ and $f''(0) = -1$.

$$log(1+x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + O(x^3) = x - \frac{1}{2}x^2 + 0(x^3)$$

$$
\begin{aligned}
logS_n(t) &= logS(0) + \tfrac{1}{2}(nt + M_{nt})log\left(\frac{\sigma}{\sqrt{n}} - \frac{\sigma^2}{2n} + O(n^{-\frac{3}{2}})\right) \\
&\quad + \tfrac{1}{2}(nt - M_{nt})log\left(\frac{\sigma}{\sqrt{n}} - \frac{\sigma^2}{2n} + O(n^{-\frac{3}{2}})\right) \\
&= logS(0) + nt\left(-\frac{\sigma^2}{2n} + O(n^{-\frac{3}{2}})\right) + M_{nt}\left(\frac{\sigma}{\sqrt{n}} + O(n^{-\frac{3}{2}}\right) \\
&= logS(0) - \tfrac{1}{2}\sigma^2 t + O(n^{-\frac{1}{2}}) + \sigma W^n(t) + O(n^{-1})W^n(t)
\end{aligned}
$$

Definition Let $(\Omega, \mathcal{F}, P)$ be a probability function, for each $\omega \in \Omega$ suppose there is a function $W(t)$ that satisfies $W(0) = 0$ and that depends on ω. The following three properties are equivalent

- For all $0 = t_0 < t_1 < \cdots < t_m$, the increments

$$W(t_1) = W(t_1) - W(t_0), ..., W(t_n) = W(t_n) - W(t_{n-1})$$

  are independent and each of these increments is normally distributed with mean and variance.
- For all $0 = t_0 < t_1 < \cdots < t_m$ the random variables $W(t_1), W(t_2), ..., W(t_m)$ are jointly normally distributed with means equal to zero and co variance matri:

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  are independent and each of these increments is normally distributed with mean and variance.
- For all $0 = t_0 < t_1 < \cdots < t_m$ the random variables $W(t_1), W(t_2), ..., W(t_m)$ are jointly normally distributed with means equal to zero and co variance matri:

Martingale Property Brownian Motion is a Martingale

$$\begin{aligned}
\mathbb{E}[W(s)|\mathcal{F}(s)] &= W(s) \\
&= \mathbb{E}[(W(t) - W(s)) + W(s)|\mathcal{F}(s)] \\
&= \mathbb{E}[(W(t) - W(s))|\mathcal{F}(s)] + \mathbb{E}[W(s)|\mathcal{F}(s)] \\
&= \mathbb{E}[(W(t) - W(s))] + W(s) \\
&= W(s)
\end{aligned}$$

$$\mathbb{E}[W(t_1) - W(t_0)] = 0 \tag{31}$$

$$Var[W(t_1) - W(t_0)] = t_1 - t_0 \tag{32}$$

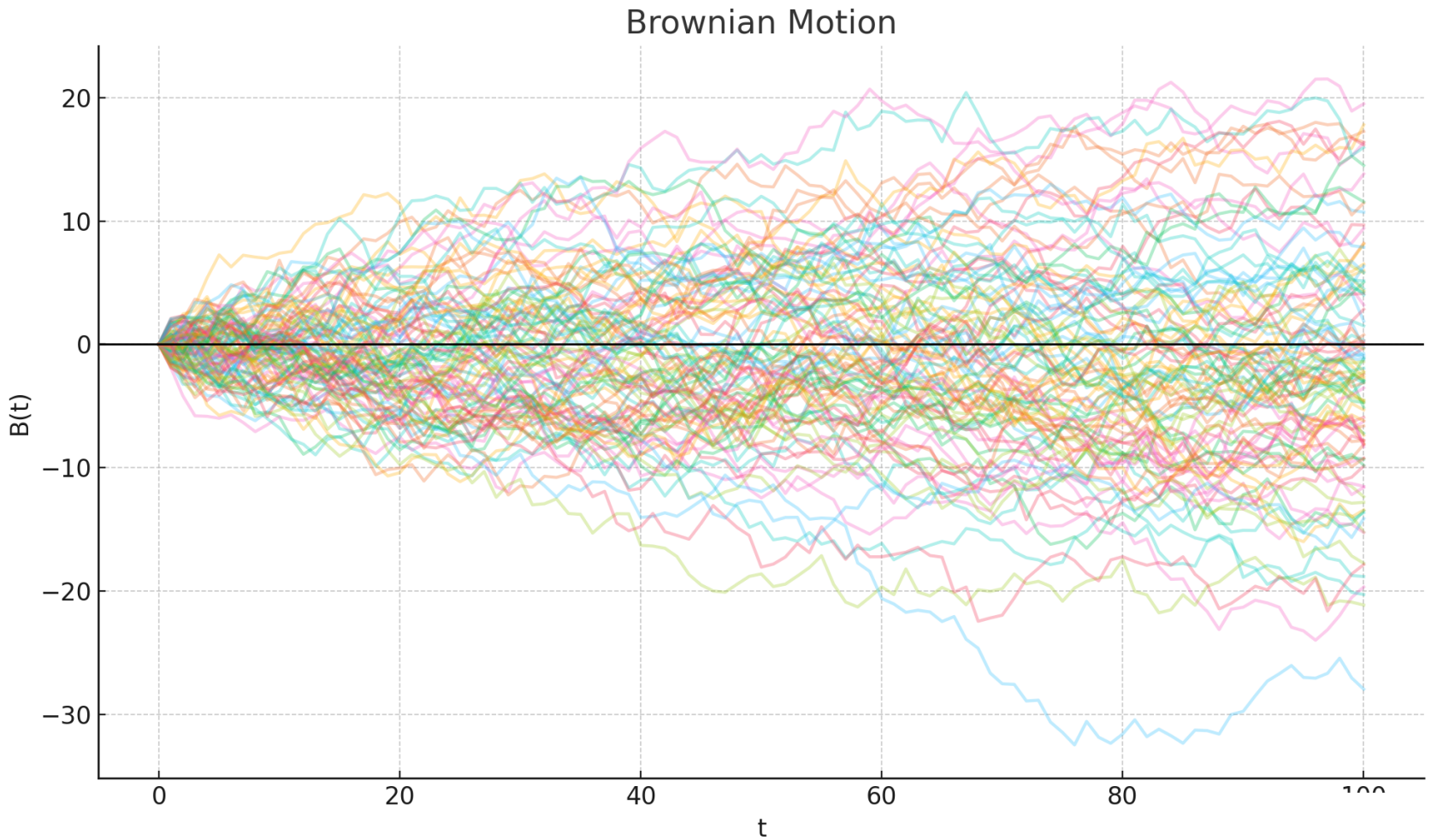
Standardized Brownian Motion

The distribution of a Brownian Motions is $W(t) \sim (0, t)$ with a standardized process Z the value for $W(t)$ is.

$$W(t) = Z\sqrt{t} = \frac{W(t) - 0}{\sqrt{t}} \tag{33}$$

Where $Z = \frac{x-\mu}{\sigma}$.

For the Geometric Brownian Motion, we use the increments of the Brownian Motion $dW(t) = \sqrt{dt}Z$


Brownian Motion
B(t)
t
20
10
0
−10
−20
−30
0
20
40
60
80

The riskless hedging principle gives the derive for the governing partial differential equation for the price of a European call option. In their seminal paper [Black and Scholes, 1973] made the following assumptions on the financial market.

- Trading takes place continuously in time.
- The riskless interest rate r is known and constant over time.
- The asset pays no dividend.
- There are no transaction costs in buying or selling the asset or the option, and no taxes.
- The assets are perfectly divisible.
- There are no penalties to short selling and the full use of proceeds is permitted.
- There are no riskless arbitrage opportunities.

Geometric Brownian Motions is: Let μ and $\sigma > 0$ adapted, a GMB is described by:

$$S(t) = S(0)e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t} \tag{34}$$

Define

$$f(t, x) = S(0)e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma x}$$

so

$$S(t) = f(t, W(t))$$

Using Taylor series

$$f_t = (\mu - \frac{1}{2}\sigma^2)f, f_x = \sigma f, f_{xx} = \sigma^2 f$$

The stochastic integral is a special integral with no solution under normal integration standards, since the random process that is used as a differential factor can be any result. It makes difficult to find a reliable result to evaluate the probability of an event.

Expected Value

$$\int_{-\infty}^{\infty} x f(x) dx \tag{35}$$

In this case, the integral is solved by the differentiation of x in dx which is a constant and allows obtaining the expected value of the integral 35 in a time interval between $0 \leq T$.

This measure is known as the Lebesgue measure, where for a space of bounded intervals in $\mathcal{B}(\mathbb{R})$, a function $F : \mathbb{R} \to \mathbb{R}$ is generated. with a $\mathcal{F}_{t\geq 0}$ filter. [Shreve, 2004]

$$\int_0^T \Delta(t)dg(t) = \int_0^T \Delta(t)g'(t)dt \tag{36}$$

$$\int_0^T \Delta(t)dW(t) \tag{37}$$

The equation 37 is a stochastic integral with $W(t)$ as Brownian Motion.

$$I(t) = \sum_{j=0}^{k-1} \Delta(t_j)[W(t_{j+1}) - W(t_j)] + \Delta(t_k)[W(t) - W$$

MartingaleThe integral Ito is a martingale the expected value given available information must exist with a $\mathcal{F}_{t\geq 0}$ filter.

$$\mathbb{E}[\Delta(t_j)W(t_{j+1}) - W(t_j)|\mathcal{F}_s] = 0 \tag{39}$$

Isometry Ito isometry allows transferring the results between the different sample spaces to the set of elements $\mathbb{R}$.

$$\mathbb{E}I^2(t) = \mathbb{E}\int_0^t \Delta^2(u)du \tag{40}$$

Quadratic Variation Using the same principle of Brownian motion, where the multiplication of two stochastic processes is equal to a squared rate of change, differentiated over a time interval.

$$[I, I] = \int_0^t \Delta^2(u)du$$

Variation integral form

$$\Delta I(t) = \int_0^t \Delta(u) dW(u) \tag{42}$$

Variation differential form

$$dI(t) = \Delta(t) dW(t) \tag{43}$$

Ito Integral with initial value 0 plus variation integral form

$$I(t) = I(0) + \int_0^t \Delta(u) dW(u) \tag{44}$$

The Taylor series, gives a solution obtain for the intervals in which the Brownian motion was being evaluated [Shreve, 2004].

$$\frac{d}{dt}f(W(t)) = f'(W(t))W'(t) \tag{45}$$

$$df(W(t)) = f'(W(t))W'(t)dt = f'(W(t))dW(t) \tag{46}$$

$$f(W(t)) = f'(W(t))dW(t) + \frac{1}{2}f''(W(t))dt \tag{47}$$

$$f(W(t)) - f(W(0)) = \int_0^t f'(W(u))dW(u) + \frac{1}{2}\int_0^t f''(W(u))du$$

$$
\begin{aligned}
f(t,W(t)) \quad = \quad & f(0,W(0)) + \int_0^T f_t(t,W(t))dt \\
& + \int_0^T f_x(t,W(t))dW(t) \\
& + \frac{1}{2}\int_0^T f_{xx}(t,W(t))dW(t)dW(t) \\
& + \frac{1}{2}\int_0^T f_{tx}(t,W(t))dtdW(t) \\
& + \frac{1}{2}\int_0^T f_{tt}(t,W(t))dtdt
\end{aligned}
$$

Ito´s Rule :

- $dW(t)dW(t) = dt$; $dtdW(t) = dW(t)dt = 0$; $dtdt = 0$

$$\begin{aligned} f(t,W(t)) &= f(0,W(0)) + \int_0^T f_t(t,W(t))dt \\ &+ \int_0^T f_x(t,W(t))dW(t) \\ &+ \frac{1}{2}\int_0^T f_{xx}(t,W(t))dW(t)dW(t) \\ &+ \frac{1}{2}\int_0^T f_{tx}(t,W(t))dtdW(t) \\ &+ \frac{1}{2}\int_0^T f_{tt}(t,W(t))dtdt \end{aligned}$$

Ito´s Rule :

- $dW(t)dW(t) = dt;\ dtdW(t) = dW(t)dt = 0;\ dtdt = 0$

$$\begin{aligned} f(t,W(t)) &= f(0,W(0)) + \int_0^T f_t(t,W(t))dt \\ &+ \int_0^T f_x(t,W(t))dW(t) \\ &+ \frac{1}{2}\int_0^T f_{xx}(t,W(t))dt \end{aligned} \tag{49}$$

Definition Let $W(t)$, $t \geq 0$, a Brownian Motion, y $\mathcal{F}(t)$ with an adapted filtration, an Itos process is [Shreve, 2004]:

$$X(t) = X(0) + \int_0^t \sigma dW(s) + \int_0^t \mu ds \tag{50}$$

Where $X(0)$ is the initial value, and σ y μ adapted. With quadratic variation $[X, X](t) = \int_0^t \sigma^2(u)du$

Ito Doeblin The Doeblin solution with the Taylor series for a stochastic process X_t is:

$$\begin{aligned} f(T, X(T)) &= f(0, X(0)) + \int_0^T f_t(t, X(t))dt + \\ &\quad \int_0^T f_x(t, X(t))\sigma dW(t) + \int_0^T f_x(t, X(t))\mu dt + \\ &\quad \frac{1}{2}\int_0^T f_{xx}(t, X(t))\sigma(t)^2 dt \end{aligned} \tag{51}$$

$$\begin{aligned} dX(t) &= \sigma dW(t) + \mu dt \\ dX(t)dX(t) &= \sigma^2(t)dW(t)dW(t) + 2\sigma\mu dW(t)dt + \mu^2(t)dtdt \\ &= \sigma^2(t)dt \end{aligned}$$

Ito´s lema Let $X(t)$ an Ito process, ethe for a stochastic process $f_t = f(X(t), t)$

$$df = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dW \tag{52}$$

Where $df = f(T, X(T)) - f(0, X(0))$
Example Let $W(t)$ a Brownina Motion with filtration $\mathcal{F}_{t\geq 0}$ and $\mu(t)$, $\sigma(t)$ an adapted process:

$$X(t) = X(0) + \int_0^t \sigma(s) dW(s) + \int_0^t (\mu(s) - \frac{1}{2}\sigma^2(s)) ds \tag{53}$$

Where $r = \mu(s) - \frac{1}{2}\sigma^2(s)$

$$dX(t) = \sigma(t)dW(t) + (\mu(t) - \frac{1}{2}\sigma^2(t))dt \tag{54}$$

$$dX(t)dX(t) = \sigma^2 dW(t)dW(t) = \sigma^2(t)dt \text{ (Quadratic Variation)}$$

$$S(t) = S(0)e^{X(t)} \tag{55}$$

$$S(t) = S(0)exp\left\{ \int_0^t \sigma(s)dW(s) + \int_0^t (\mu(s) - \frac{1}{2}\sigma^2(s))ds \right\} \tag{56}$$

$$\begin{aligned} dS(t) &= df(X(t)) \\ &= f'(X(t))dX(t) + \tfrac{1}{2}f''(X(t))dX(t)dX(t) \\ &= S(0)e^{X(t)}dX(t) + \tfrac{1}{2}S(0)e^{X(t)}dX(t)dX \\ &= S(t)\mu dt + \sigma(t)S(t)dW(t) \end{aligned} \tag{57}$$

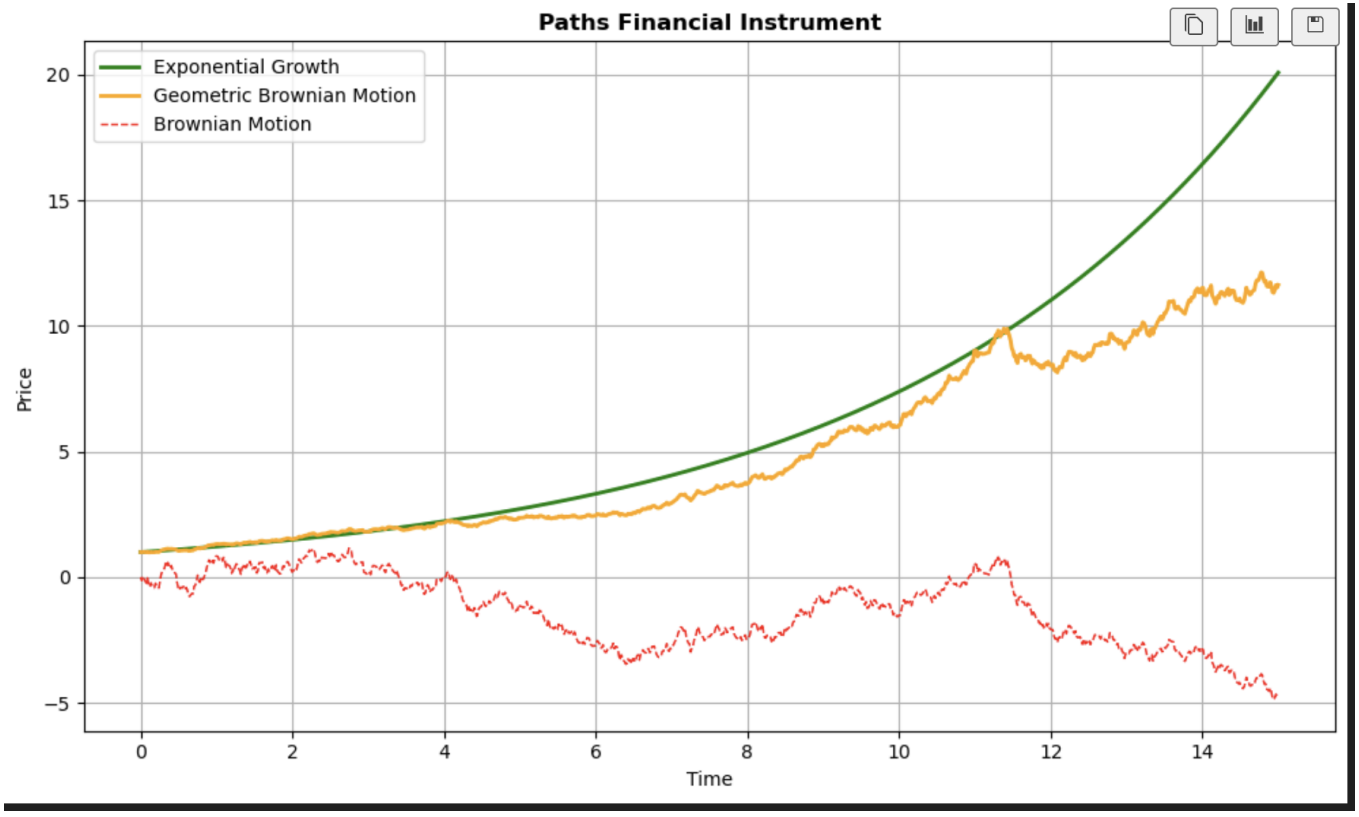
Geometric Brownian motion in differential form is

$$dS(t) = \mu S(t)dt + \sigma S(t)dW(t) \tag{58}$$

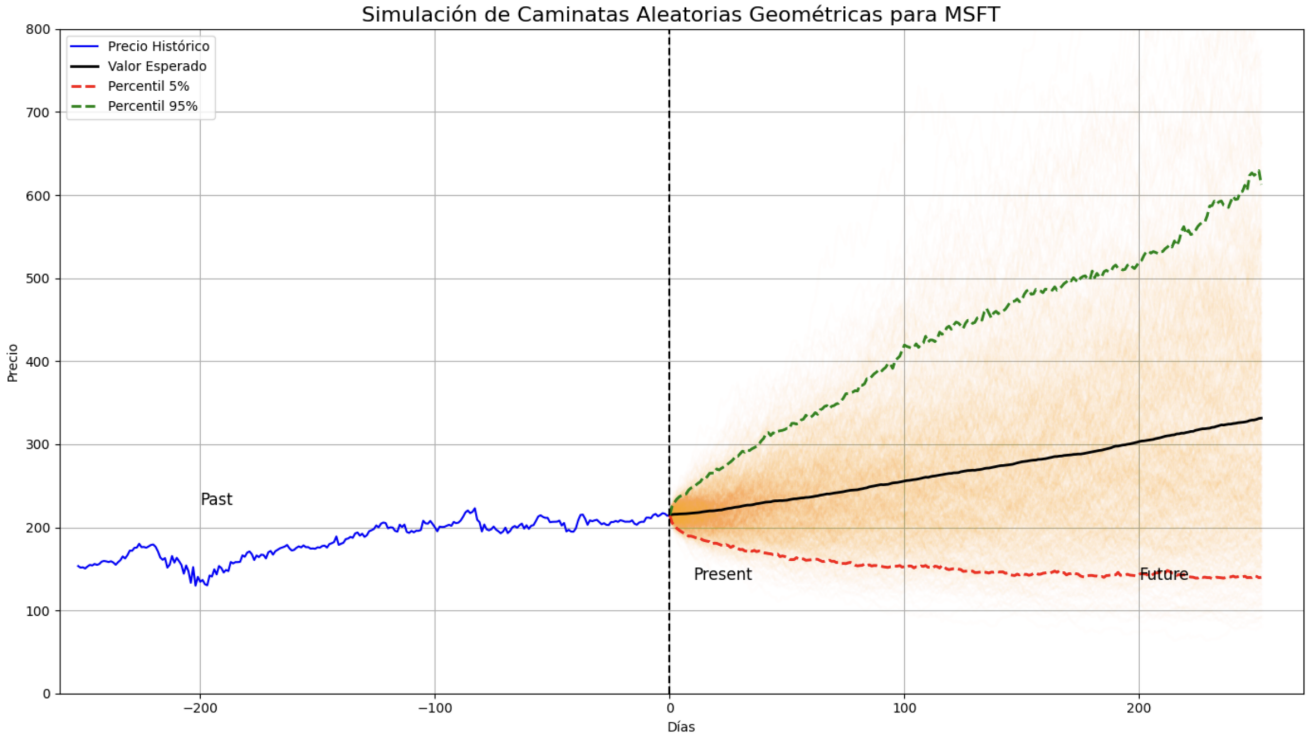
The Integral form is:

$$S(t) = S(0)exp\left\{ \int_0^t \sigma(s)dW(s) + \int_0^t (\mu(s) - \frac{1}{2}\sigma^2(s))ds \right\} \tag{59}$$

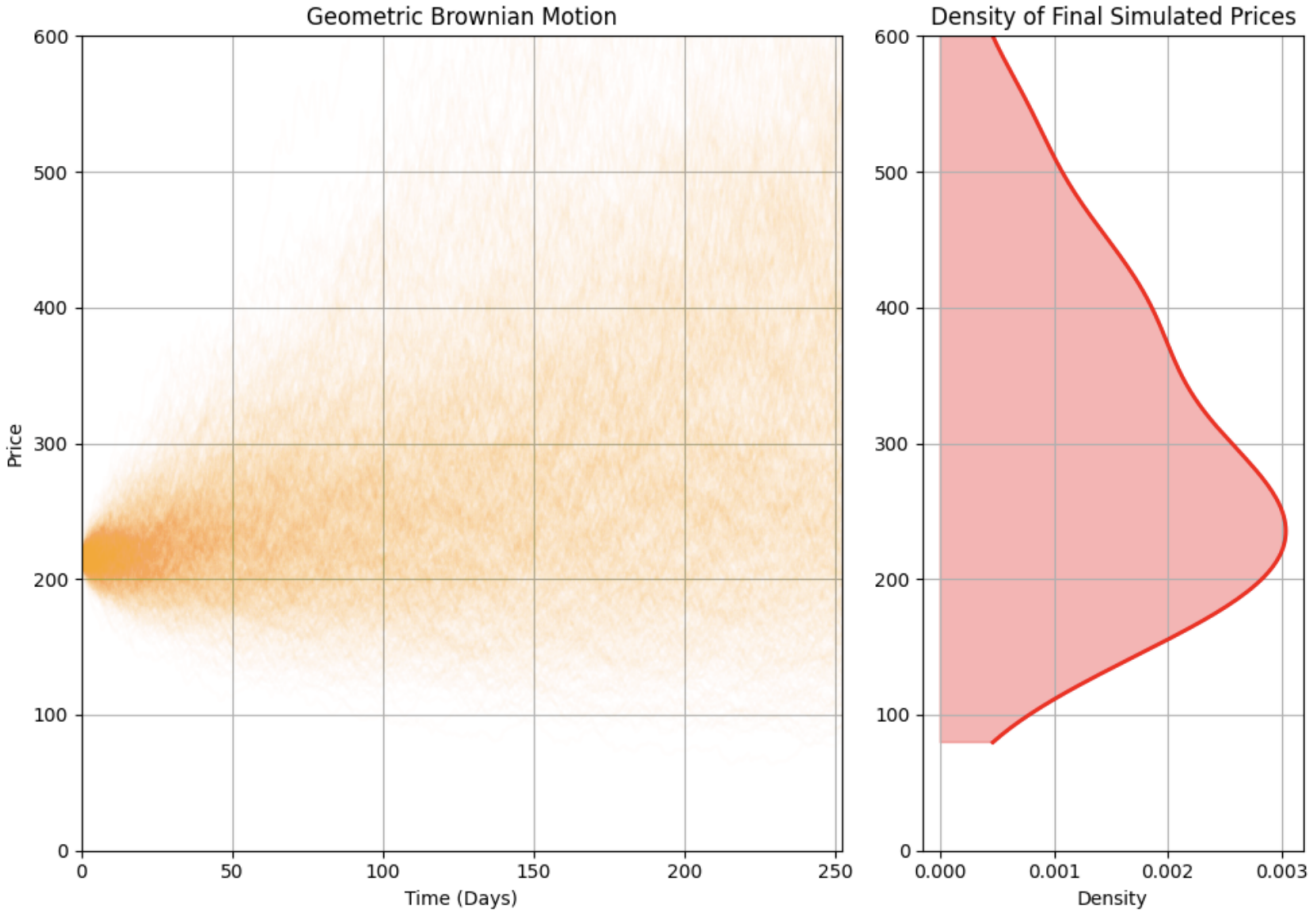
Exponential Grow



Forecast


Simulación de Caminatas Aleatorias Geométricas para MSFT
Precio Histórico
Valor Esperado
Percentil 5%
Percentil 95%
Precio
Días
Past
Present
Future
800
700
600
500
400
300
200
100
0
−200
−100
0
100
200

Density


Geometric Brownian Motion
Price
Time (Days)
0
50
100
150
200
250
0
100
200
300
400
500
600
Density of Final Simulated Prices
Density
0.000
0.001
0.002
0.003

Simulation Process: Euler Maruyama (Montecarlo Simulation)

$S(t) = S(t-1) + dS(t)$ and $W(t) = \sqrt{t}Z$

$$
\begin{aligned}
S(0) &= S(0) \\
S(1) &= S(0) + \mu S(0)dt + S(0)\sigma\sqrt{dt}Z \\
S(2) &= S(1) + \mu S(1)dt + S(1)\sigma\sqrt{dt}Z \\
S(3) &= S(2) + \mu S(2)dt + S(2)\sigma\sqrt{dt}Z \\
S(4) &= S(3) + \mu S(3)dt + S(3)\sigma\sqrt{dt}Z \\
S(5) &= S(4) + \mu S(4)dt + S(4)\sigma\sqrt{dt}Z \\
S(t) &= S(t-1) + \mu S(t-1)dt + S(t-1)\sigma\sqrt{dt}Z
\end{aligned}
$$

An option gives the holder the right (but not the obligation) to buy or sell an asset by a certain date for a pre- determined price, is classified either as a call option or a put option. A call (or put) option is a contract which gives its holder the right to buy (or sell) a prescribed asset, known as the underlying asset, by a certain date (expiration date) for a predetermined price.

If the option can only be exercised on the expiration date, then the option is called a European option. Otherwise, if the exercise is allowed at any time prior to the expiration date, then the option is called an American option.

The counterparty to the holder of the option contract is called the option writer. The holder and writer are said to be, respectively, in the long and short positions of the op contract.

Factors

- Current Spot Price of the underlying S_0
- Strike price K
- Time to expiration T
- The volatility of the underlying σ
- The risk free rate r

Pay-off of the option

- Call $V_{call} = max\{S - K, 0\}$ Put $V_{put} = max\{K - S, 0\}$

Moneyness

- in-the-money if $S > K$ for call or $S < K$ for put
- out of the money if if $K > S$ for call or $K < S$ for put
- At the money if $S = K$

Black, Scholes and Merton model

Black and Scholes (1973) revolutionized the pricing theory of options showing how to hedge continuously the exposure on the short position of an option. Consider the writer of a European call option on a risky asset.

He or she is exposed to the risk of unlimited liability if the asset price rises above the strike price. To protect the writer's short position in the call option, he or she should consider purchasing a certain amount of the underlying asset so that the loss in the short position in the call option is offset by the long position in the asset. In this way, the writer is adopting the hedging procedure.

Self- Financing Strategy: Suppose an investor holds a portfolio of securities, such as a combination of options, stocks and bonds. As time passes, the value of the portfolio changes because the prices of the securities change. Besides, the trading strategy of the investor affects the portfolio value by changing the proportions of the securities held in the portfolio, say, and adding or withdrawing funds from the portfolio.

An investment strategy is said to be self-financing if no extra funds are added or withdrawn from the initial investment. The cost of acquiring more units of one security in the portfolio is completely financed by the sale of some units of other securities within the same portfolio.

Evolution of a Portfolio to determine the necessary capital in a hedging operation using options. Given a portfolio with a value of $Y(t)$, it is invested with an interest rate r, rhat follows a geometric Brownian motion 57 which is the solution of 59

$dY(t) = rY(t)dt$ is the benefit of the portfolio.

Assumptions

- The portfolio Y_t is riskless
- The portfolio is self financed
- No dividens
- No arbitrage
- It is allow short positions

$$dY_t = a_t dB(t) + c_t dS(t) - dV(t, S(t)) \tag{60}$$

Where a_t and c_t are units of $B(t)$ and $S(t)$.

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dW \tag{61}$$

$$\begin{aligned} dY &= \left[a_t r B + c_t \mu S - \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) \right] dt \\ &\quad + \left(c_t \sigma S - \sigma S \frac{\partial V}{\partial S} \right) dW \end{aligned}$$

For $c_t = \frac{\partial V}{\partial S}(S(t), t)$

$$\begin{aligned} r\left(a_t B + \frac{\partial V}{\partial S} S - V\right) dt &= \left(a_t r B + c_t \mu S - \frac{\partial V}{\partial t} \right. \\ &\quad \left. -\mu S \frac{\partial V}{\partial t} - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right) dt \\ r\left(a_t B + \frac{\partial V}{\partial S} S - V\right) dt &= \left(a_t r B - \frac{\partial V}{\partial t} - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right) dt \end{aligned} \tag{63}$$

$$\frac{\partial V}{\partial t} + \frac{\partial V}{\partial S} r S + \frac{1}{2}\frac{\partial^2 V}{\partial S^2} S^2 \sigma^2 - rV = 0 \tag{64}$$

Partial Differential Equation Black and Scholes

The governing equation for a European put option can be derived similarly and the same Black–Scholes equation is obtained. Let V (S, t) denote the price of a derivative security with dependence on S and t, it can be shown that V is governed by

$$\frac{\partial V}{\partial t} + \frac{\partial V}{\partial S} rS + \frac{1}{2}\frac{\partial^2 V}{\partial S^2} S^2\sigma^2 - rV = 0 \tag{65}$$

These solutions gives the fundamental steps to evaluate any financial instrument

Pricing an European Call

$$V(S,t) = SN(d1) - Ke^{-r(T-t)}N(d2) \tag{66}$$

We calculated the expected value of the current price of an option.

$$\mathbb{E}\left[e^{-rT}(S(T)-K)^{+}\right] = S(0)N(d_1(T,S(0))) - Ke^{-rT}N(d_2(T,S(0)))$$

Pricing an European Call

$$V(S,t) = SN(d1) - Ke^{-r(T-t)}N(d2) \tag{67}$$

We calculated the expected value of the current price of an option.

$$\mathbb{E}\left[e^{-rT}(S(T)-K)^{+}\right] = S(0)N(d_1(T,S(0))) - Ke^{-rT}N(d_2(T,S(0)))$$

Where

$$d_{1,2}(T,S(0)) = \frac{1}{\sigma\sqrt{T}}\left[\log\frac{S(0)}{K} + \left(r \pm \frac{\sigma^2}{2}\right)T\right],$$

N is the Normal CDF

$$N(y) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{y} e^{-\frac{1}{2}z^2}dz = \frac{1}{\sqrt{2\pi}}\int_{-y}^{\infty} e^{-\frac{1}{2}z^2}$$

$$
\begin{aligned}
& \mathbb{E}\left[e^{-rT}(S_T-K)^+\right] \\
= \; & e^{-rT}\int_{\frac{1}{\sigma}\left[\ln\frac{K}{S_0}-(r-\frac{1}{2}\sigma^2)T\right]}^{\infty}\left(S_0e^{(r-\frac{1}{2}\sigma^2)T+\sigma x}-K\right)\frac{e^{-\frac{x^2}{2T}}}{\sqrt{2\pi T}}dx \\
= \; & e^{-rT}\int_{\frac{1}{\sigma\sqrt{T}}\left[\ln\frac{K}{S_0}-(r-\frac{1}{2}\sigma^2)T\right]}^{\infty}\left(S_0e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}y}-K\right)\frac{e^{-\frac{y^2}{2}}}{\sqrt{2\pi}}dy \\
= \; & S_0e^{-\frac{1}{2}\sigma^2T}\int_{\frac{1}{\sigma\sqrt{T}}\left[\ln\frac{K}{S_0}-(r-\frac{1}{2}\sigma^2)T\right]}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{y^2}{2}+\sigma\sqrt{T}y}dy \\
& -Ke^{-rT}\int_{\frac{1}{\sigma\sqrt{T}}\left[\ln\frac{K}{S_0}-(r-\frac{1}{2}\sigma^2)T\right]}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{y^2}{2}}dy \\
= \; & S_0\int_{\frac{1}{\sigma\sqrt{T}}\left[\ln\frac{K}{S_0}-(r-\frac{1}{2}\sigma^2)T\right]-\sigma\sqrt{T}}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{\xi^2}{2}}d\xi \\
& -Ke^{-rT}N\left(\frac{1}{\sigma\sqrt{T}}\left(\ln\frac{S_0}{K}+(r-\tfrac{1}{2}\sigma^2)T\right)\right) \\
= \; & S_0N(d_1(T,S_0))-Ke^{-rT}N(d_2(T,S_0)).
\end{aligned}
$$

Put-Call Parity In general, after a result is obtained for a European call option, the corresponding result for a European put option can be derived by using the Put-Call parity.

Put-call parity states the relation between the prices of a pair of call and put options. For a pair of European put and call options on the same underlying asset and with the same expiration date and strike price.

Put Call Parity

$$
\begin{aligned}
P(S,t) &= Ke^{-r(T-t)} - Se^{-q(T-t)} + C(S,t) \\
&= Ke^{-r(T-t)} - Se^{-q(T-t)} \\
&\quad +(Se^{-q(T-t)}N(d1) - Ke^{-r(T-t)}N(d2) \\
&= Ke^{-r(T-t)}(1 - N(d2)) - Se^{-q(T-t)}(1 - N(d1)) \\
&= Ke^{-r(T-t)}N(-d2) - Se^{-q(T-t)}N(-d1)
\end{aligned}
$$

$$
\begin{aligned}
P(S,t) &= Ke^{-r(T-t)} - Se^{-q(T-t)} + C(S,t) \\
&= Ke^{-r(T-t)} - Se^{-q(T-t)} \\
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&= Ke^{-r(T-t)}(1 - N(d2)) - Se^{-q(T-t)}(1 - N(d1)) \\
&= Ke^{-r(T-t)}N(-d2) - Se^{-q(T-t)}N(-d1)
\end{aligned}
$$

Remarks

- The original derivation of the governing partial differential equation focuses on the financial notion of riskless hedging but misses the precise analysis of the dynamic change in the value of the hedged portfolio.
- The ability to construct a perfectly hedged portfolio relies on the assumption of continuous trading and continuous asset price path. These two and other assumptions in the Black–Scholes pricing model have been critically examined by later works in derivative pricing theory.
- The interest rate is widely recognized to be fluctuating over time in an irregular manner rather than being constant.
- The Black–Scholes pricing approach assumes continuous hedging at all times. In the real world of trading transaction costs.

Thank you

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library(cowplot) library(ggplot2)
p2 ¡- ggdraw() +
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1,0)$p2$
@

Trading Strategies :A spread strategy refers to a portfolio which consists of options of the same type (that is, two or more calls, or two or more puts) with some options in the long position and others in the short position in order to achieve a certain level of hedging effect.

- Bull (Long Strategy)
- Bear (Short Strategy)
- Straddle
- Strangle
- Butterfly

Bull (Long Strategy) :Is a long position because is created by buying a call option on an underlying with a certain strike price and selling an option on the same stock with a higher strike price.

Event	**Call**	**S-Call**	**Pay-off**
$S < K_1$	0	0	0
$K_1 < S < K_2$	$S - K_1$	0	$S - K_1$
$S > K_2$	$S - K_1$	$K_2 - S$	$K_2 - K_1$

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Bear (Short Strategy) :Is a short position because is created by buying a put option on an underlying with a certain strike price and selling an option on the same stock with a lower strike price.

Event	**Put**	**S-Put**	**Pay-off**
$S < K_1$	$K_2 - S$	$S - K_1$	$K_2 - K_1$
$K_1 < S < K_2$	$K_2 - S$	0	$K_2 - S$
$S > K_2$	0	0	0

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Straddle :Involves buying a call option on an underlying with a certain strike price and buying a put option on the same stock.

Event	**Put**	**Call**	**Pay-off**
$S < K$	$K - S$	0	$K - S$
$S > K$	0	$S - K$	$S - K$

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1,0)$p2$
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Strangle :Involves buying a call option on an underlying with a certain strike price and buying a put option on the same stock with lower strike price.

Event	**Put**	**Call**	**Pay-off**
$S < K_1$	$K_1 - S$	0	$K_1 - S$
$K_1 < S < K_2$	0	0	0
$S > K_2$	0	$S - K_2$	$S - K_2$

¡¡echo=FALSE,eval=TRUE,fig=true,width=15,height=15 ¿¿=
library(cowplot) library(ggplot2)
p2 ¡- ggdraw() +
draw$_i mage(”/Users/andresmartinez/Dropbox/Uexternado/Maestr$
1,0)$p2$
@

Butterfly :Uses four options wich two of them are short positions with the same strike price, and two long call with different strike price.

Event	**Call** K_1	**2S-Call** K_2	**Call** K_3	**Pay-off**
$S < K_1$	0	0	0	0
$K_1 < S < K_2$	$S - K_1$	0	0	$S - K_1$
$K_2 < S < K_3$	$S - K_1$	$2(K_2 - S)$	0	$2K_2 - K_1 - S$
$S > K_3$	$S - K_1$	$2(K_2 - S)$	$S - K_3$	$2K_2 - K_3 - K$

¡¡echo=FALSE,eval=TRUE,fig=true,width=15,height=15 ¿¿=
library(cowplot) library(ggplot2)
p2 ¡- ggdraw() +
draw$_i mage(”/Users/andresmartinez/Dropbox/Uexternado/Maestr$
1,0)$p2$
@

Summary

Call		Put	
Buy		Sell	
Payoff	K Strike Price	Payoff	
$V(S,t) =$	$max\{S-K,0\}$	$V(S,t) =$	$max\{K-S,0\}$
Si $S < K$	$V(S,t) = 0$	Si $S < K$	$V(S,t) = K - S$
Si $S > K$	$V(S,t) = S - K$	Si $S > K$	$V(S,t) = 0$
Both with expiration $0 \leq t \leq T$			

Un bono cero cupón con madurez T es un instrumento financiero que paga un valor de una unida al final del plazo, el precio del bono se descuenta en t como $B(t,T)$ para $t \in [0,T]$.

$$B(t,T) = exp(-r(T-t)) \tag{68}$$

El precio de un bono $B(t,T)$ está definido como una variable aleatoria en un espacio muestral $(\Omega, \mathcal{F}, \mathbb{P})$. Siendo $B(t,T)$ discreto con filtración $\mathcal{F}_t$.

Mientras que el precio de un bono es igual a 1 en el momento de su maduración $B(T,T) = 1$.
(G_t) Es el factor de descuento que al momento 0 de la inversión es $G(0,T) = 1$ y van aumentando de forma geométrica a medida que cambia la tasa de interés en cada período,

$$G_t = (1 + r(t))$$

G_t Es libre de arbitraje y existe para $\mathbb{P}$ una medida de riesgo neutral $\mathbb{Q}$, por lo tanto es una martingala $\frac{B(t,T)}{G(t,T)}$ o con una filtración $\mathcal{F}_t$.

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$$\frac{B(t,T)}{G(t,T)} = \mathbb{E}_{\mathbb{Q}}\left(\frac{B(t,\tau)}{G(t,\tau)}|\mathcal{F}_t\right) \quad para \quad 0 \leq t \leq \tau < T$$

Un bono con dos años de maduración, tasa de interés inicial de $r_0 = 6\,\%$ y valor inicial $B(0,2) = 0{,}92$, puede variar un $1\,\%$ en el período 1, si la ganancia inicial es $G_0 = 1$. Revise el supuesto de arbitraje y martingala usando un espacio muestral $\Omega = \{\omega_1, \omega_2\}$.

Solución: El objetivo es ver si existe una posibilidad de arbitraje. La información se obtiene con $\mathcal{F}_{t\geq 0}$. En este caso se puede hacer operaciones en largo (comprar barato y vender caro) y en corto que significa vender un producto sin tenerlo para recomprarlo después más barato. Se puede invertir o pedir prestado a la tasa libre de riesgo vigente de cada uno de los períodos. En caso de que exista una posibilidad de arbitraje, la idea es establecer que estrategia se puede realizar para generar ganancia sin riesgo.

Para $\mathcal{F}_2$ como conjunto potencia de Ω, se tiene $\mathcal{F}_0 = \{\emptyset, \Omega\}$ y $\mathcal{F}_1 = \{\emptyset, \Omega\, \omega_1, \omega_2\}$. Siendo $r_0 = 6\,\%$ y dados se obtiene r_1, $G_2(\omega)$ y $B(1,2)$ en la siguiente tabla.

ω	$r(\omega)$	$G_2(\omega)$	$B(1,2)$
ω_1	$5\,\%$	1,1130	0,9524
ω_2	$7\,\%$	1,1342	0,9346

Donde $B(2,2) = 1$, $G_0 = 1$ y $G_1 = 1{,}06$

$$\frac{B(0,2)}{G(0,2)} = 0{,}92 = \mathbb{E}_{\mathbb{Q}}\left(\frac{B(1,2)}{G(1,2)}|\mathcal{F}_0\right) = \mathbb{E}_{\mathbb{Q}}\left(\frac{B(1,2)}{1{,}06}|\mathcal{F}_0\right)$$

Martingala $q = \mathbb{Q}(\omega_1) + \mathbb{Q}(\omega_3)$

$$0{,}92 = q\frac{0{,}9524}{1{,}06} + (1-q)\frac{0{,}9346}{1{,}06} = 0{,}88017 + 0{,}0168q$$

El resultado es $q = 2{,}2798$, dado que el intervalo de la probabilidad está entre $0 \leq q \leq 1$ en este caso hay arbitraje y no existe probabilidad de riesgo neutral.

Como hay arbitraje lo más lógico es revisar la estrategia que se debería emplear para generar ganancias a cero riesgo.

Período	$t = 0$	$t = 1$
Estrategia	Corto en Bonos(Venta)	Cierra(Compra)
Invierte	0,92 al 6 %	
Recibe		$1{,}06 * 0{,}9752$
Escenario 1	La tasa baja	$1{,}05 * 0{,}92 = 0{,}9524$
Escenario 2	La tasa sube	$1{,}07 * 0{,}92 = 0{,}9346$

Esto quiere decir que no importa el valor que el bono tome en el primer período, ya que el valor que se obtiene al invertir 0,92 en el período cero a una tasa libre de riesgo alcanza a cubrir cualquier variación de la tasa en el período uno.

Two concerns:

- online education
- long term stability

Country\Year	2000	2020	2030
Colombia	24	32	36
USA	35	38	40
World	26	30	33

Cuadro: Average Age by Country. Source: Our world in data

Why go to the University?

- Steady Income
- Better market conditions
- Career

Which University?

- High Income
- Public
- Low Income

Legitimacy of an Institution

[Hannan and Freeman, 1977], legitimacy is an important factor that universities have used to shape the environment and the social interactions.

Longevity of the institutions [Hannan and Freeman, 1984] and the fact that during their lifetime they develop relationships with key actors give them the opportunity to survive despite the changes in the environment.

Environment where the universities are located right now shows that there are other ways to access to high education with more flexibility, so the students can manage better their time between academic activities, work and personal life (travel, family, etc).

Job market demands more technical and digital skilled workers, fields in which virtual education can be more efficient than traditional universities

There are few spaces to research activities or even worse, research is not considered necessary for all levels of the organization.

How long the universities in the actual environment could continue satisfy or create contributions to the society?

The structure presented in the work from [Carroll, 1984] begins with some reflections about how ecology theory has evolved to explain the change and the relation between the organization and environment

[Hannan and Freeman, 1977] expose the symbiotic relations between social units to theories that emphasize competitive relations and also compare theories that assume temporal equilibrium with theories that explicitly emphasize disequilibrium.

In those works, the authors argument also that organizations have changes and mortality, mergers or acquisitions as result of the environmental context no matter their size, but the problem is how the environmental forces change the organizat structure and how they maintain the reliability.

The arguments mentioned before can be summarized in three levels:

- **Developmental Approach to Evolution** Organizations are the results of the environment.
- **Selection Approach to Evolution** This argument is developed also by [Carroll, 1984] using three stages from [Campbell, 1965]:
  - Variation
  - Selection
  - Retention
- **Macro-evolutionary Approach**: Focused on the community of the organizations

This empirical study proposes to compute the death rate or probability of an university as well as the average time in which this scenario can happen according to the interactions of educational institutions with their environment.

The author [Carroll, 1981] construct a model which measures the capacity of the primary PRI, secondary SEC and tertiary TER organizations. The last one is focus on graduate and undergraduate segments.

$$1/TER = c_0 + c_1(1/R) + c_2(1/R^2) \tag{69}$$

$$\frac{\partial TER}{\partial t} = f(PRI, SEC, TER) \tag{70}$$

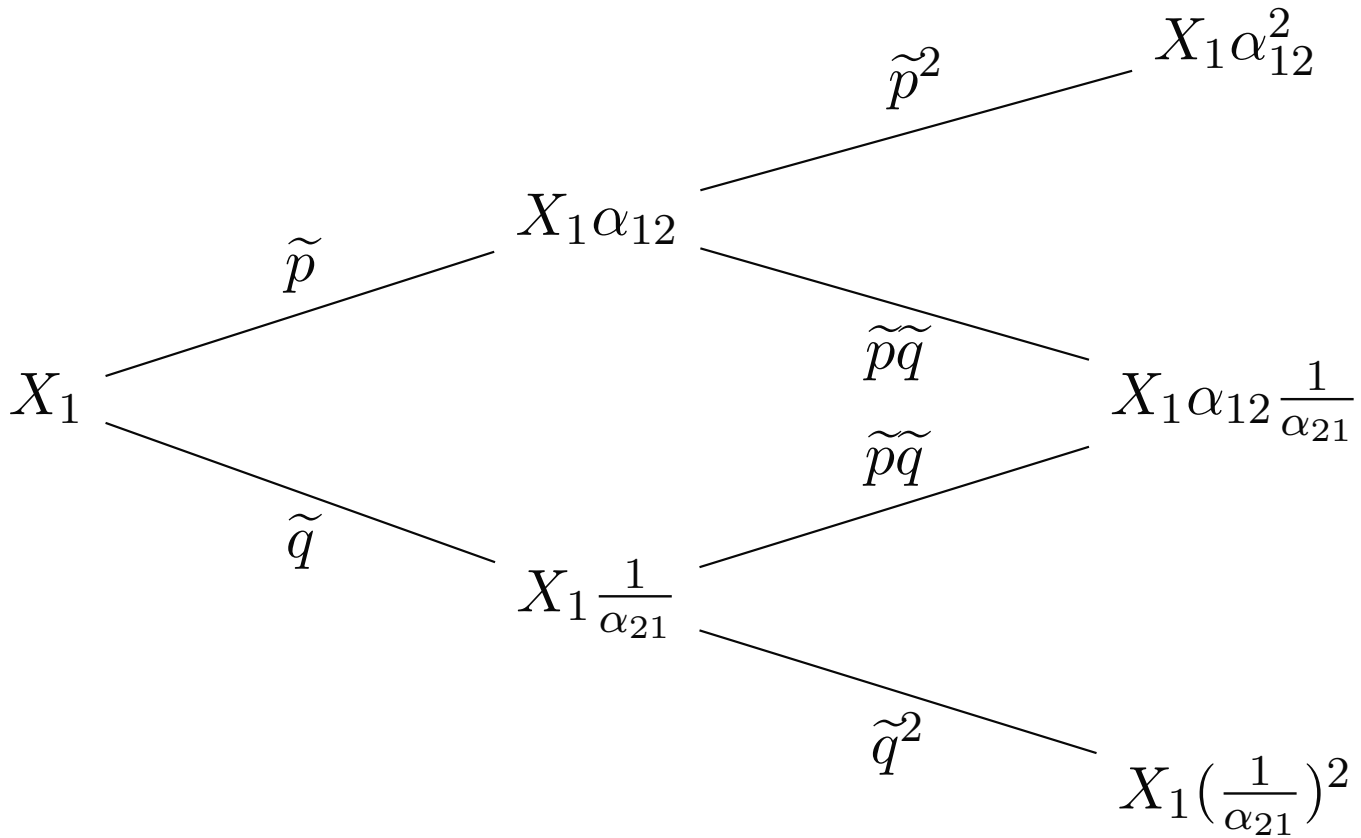
The competitive dynamic is α_i where α_1 is for high income, α_2 is for public universities, α_3 is for low income and α_4 is for other kind of institutions is defined as:

- $\alpha_1 = \beta_0 + \beta_1 \frac{\partial TER}{\partial t} + \beta_2 HI_t + \beta_3 ES_t + \beta_4 Scores + \beta_5 \alpha_2 + \beta_6 \alpha_3 + \beta_7 \alpha_4 + \epsilon_t$
- $\alpha_2 = \beta_0 + \beta_1 \frac{\partial TER}{\partial t} + \beta_3 ES_t + \beta_4 Scores + \beta_5 \alpha_1 + \beta_6 \alpha_3 + \beta_7 \alpha_4 + \epsilon_t$
- $\alpha_3 = \beta_0 + \beta_1 \frac{\partial TER}{\partial t} + \beta_2 LI_t + \beta_3 ES_t + \beta_4 Scores + \beta_5 \alpha_2 + \beta_6 \alpha_1 + \beta_7 \alpha_4 + \epsilon_t$
- $\alpha_4 = \beta_0 + \beta_1 \frac{\partial TER}{\partial t} + \beta_3 ES_t + \beta_5 \alpha_2 + \beta_6 \alpha_3 + \beta_7$

According to [Hannan and Freeman, 1977] selection eliminates the weakest competitors in which the Hazard function compute the death rate of an organization, so for this comparison, first it is use the equation of Lotka Volterra under the inequality $\frac{1}{\alpha_{i1}} < \frac{k_i}{k_1} < \alpha_{1i}$:

$$\frac{dX_1}{dt} = r_1 X_1 \left(\frac{k_1 - X_1 - \alpha_{12} X_1}{k_1} \right) \tag{71}$$

$$\frac{dX_2}{dt} = r_2 X_2 \left(\frac{k_2 - X_2 - \alpha_{21} X_2}{k_2} \right) \tag{72}$$


X_1
$\widetilde{p}$
$\widetilde{q}$
$X_1\alpha_{12}$
$X_1\frac{1}{\alpha_{21}}$
$\widetilde{p}^2$
$\widetilde{p}\widetilde{q}$
$\widetilde{p}\widetilde{q}$
$\widetilde{q}^2$
$X_1\alpha_{12}^2$
$X_1\alpha_{12}\frac{1}{\alpha_{21}}$
$X_1(\frac{1}{\alpha_{21}})^2$

$$G(t) = exp[-\int_0^t h(s)ds]. \tag{73}$$

$$h(t) = -\frac{dlogG(t)}{dt} \tag{74}$$

$$G(t) = \alpha_0 + \omega_0^{(\gamma_0 t)} \tag{75}$$

Be $h(t)$ a mean reversion model, means that a rate at a organization decrease is:

$$dh(t) = \gamma(\alpha - h(t))dt + \omega dW(t) \tag{76}$$

The process in summarized in the next bullets:

- Compute the population for tertiary education
- Get the competitive factor α
- Establish a binomial model with the Lotka Volterra equation.
- Compute p and q to get the survival possibility of X_i.
- Use the competitive factor compute in the discrete model to the continuous model
- Get γ and ω from the regression process of each α
- Build the stochastic model for $G(t)$ with $h(t)$ as hazard rate.

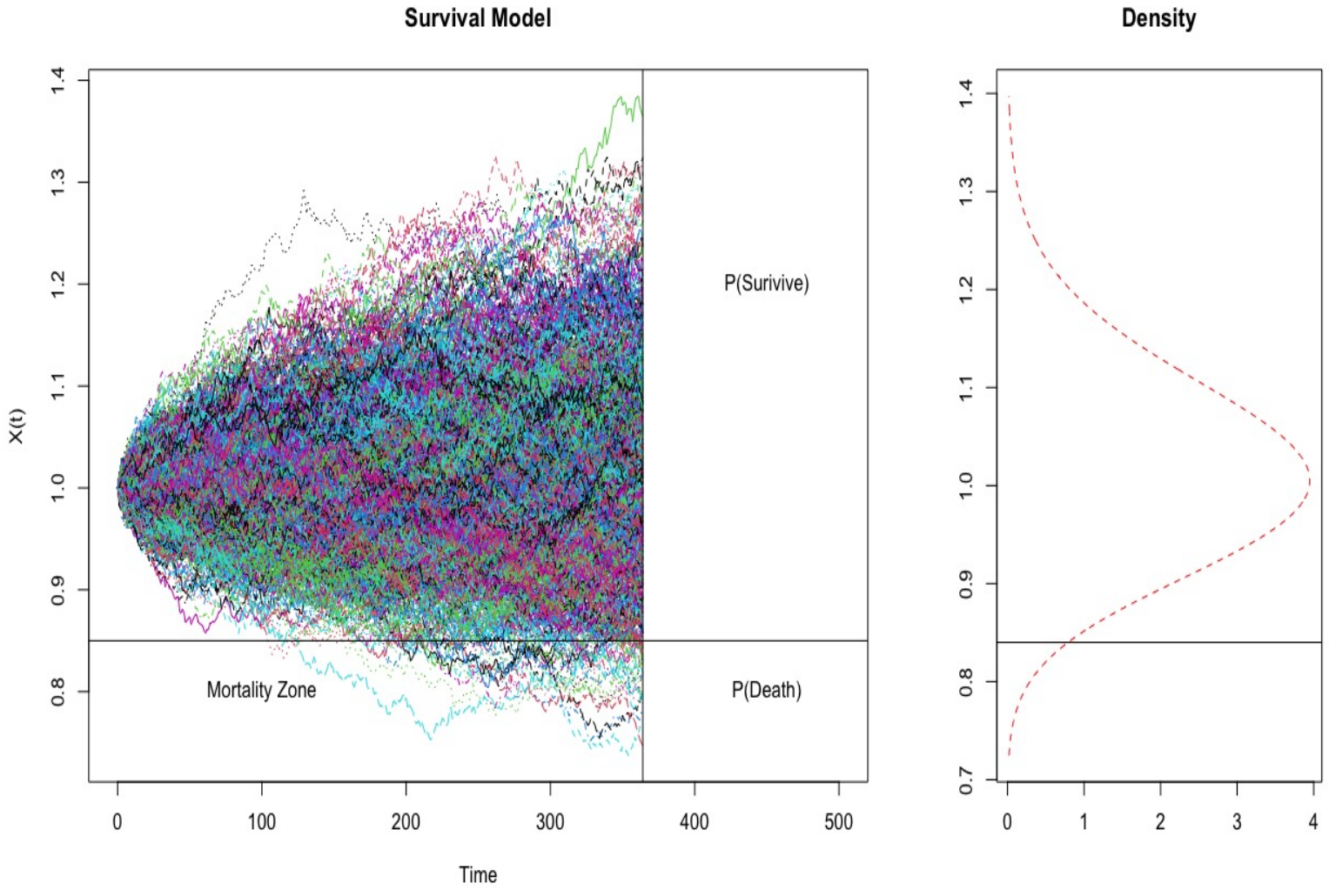


Figura: Survival rate model

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